Discriminating Underground Nuclear Explosions from Natural Earthquakes

Using Signal Processing and Machine Learning Methods

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Motivating Time-Frequency Representations

Raw amplitudes overlap; time-varying frequency content provides the primary statistical discriminant.

1. Amplitude Ambiguity

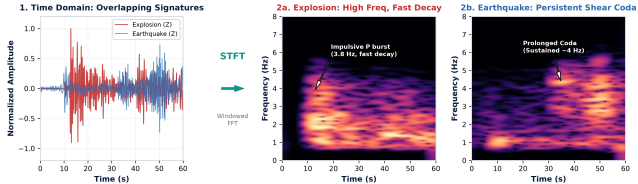
1D waveforms overlap in amplitude and envelope; magnitude alone cannot classify the source.

2. Spectral Evolution

Blasts concentrate high frequencies early; earthquakes sustain lower frequencies in late coda.

3. 2D Tensor Mapping

STFT maps 1D temporal series into 2D time-frequency grids for feature extraction and CNNs.



Workflow: 3-Component Waveform (3×1200) $\xrightarrow{\text{STFT}}$ 2D Spectrogram Tensors $(3 \times 128 \times 128)$ $\rightarrow$ Classifiers

SECTION 01

Signal Characteristics & Data Curation

Understanding source differences and constructing a leakage-free benchmark

Seismic Datasets: Sources, Features & Test Sites

Multi-station recordings, engineered feature representations, and external test sites.

Data Sources & Locations

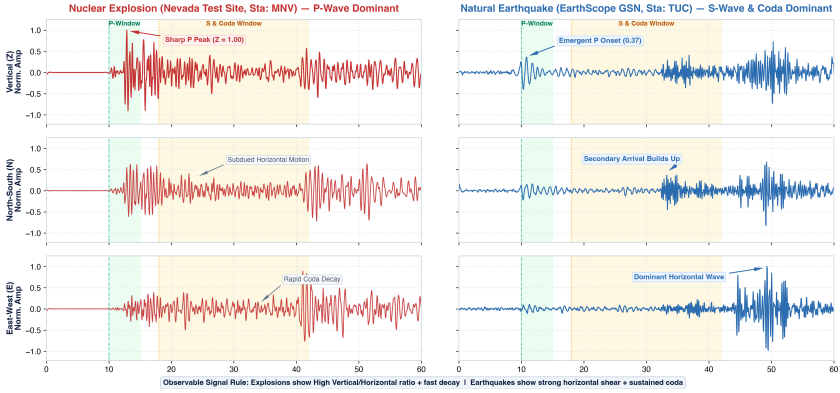
- **Explosions (NTS, USA):** 198 recordings across 67 underground nuclear tests at Nevada Test Site (LLNL GTUNE, 1968–1992).
- **Earthquakes (Global/Regional):** 150 recordings across 45 tectonic events ($M_w 5.2 - 6.2$) on EarthScope GSN stations.
- **Unseen Test Site (DPRK):** All 6 declared North Korean underground tests (2006–2017) at Punggye-ri recorded on regional sensors (MDJ, INCN).

Signal Features & Representations

- **Raw Waveforms:** Synchronized 60 s window @ 20 Hz across Z, N, E components ($3 \times 1,200 = 3,600$ values).
- **Track A (18 Physical Features):** Spectral energy ratios (P/S), Z/H energy ratios, coda decay (τ), and 3D rectilinearity.
- **Track B (2D Spectrogram Tensors):** $3 \times 128 \times 128$ time-frequency images extracted via STFT across components.

Evaluation Standard: *Station coordinates purged; cross-validated strictly by event ID and held out against unseen North Korean tests.*

Raw Waveforms: Contrasting Time-Domain Signatures



Observed Signatures: *Explosions display sharp vertical P-waves and fast decay; earthquakes show prominent horizontal shear waves and sustained coda.*

Observable Differences: Explosions vs. Earthquakes

Measurable signal properties distinguishing the two event classes across 3-component recordings.

Underground Explosion (Blast)

- **Impulsive Vertical Spike:** Sharp initial arrival predominantly on the vertical (Z) channel.
- **Horizontal Deficit:** Minimal motion detected on North (N) and East (E) sensors.
- **Early High-Frequency Burst:** Spectral energy (> 3 Hz) concentrated at onset.
- **Fast Coda Decay:** Signal returns rapidly to pre-event baseline levels.

Natural Earthquake (Tectonic Event)

- **Emergent Onset:** Amplitude builds gradually over several seconds.
- **Intense Horizontal Shaking:** Dominant motion on North (N) and East (E) sensors.
- **Persistent Low Frequencies:** Spectral power concentrated in low-to-mid bands (< 3 Hz).
- **Prolonged Coda Duration:** Reverberating late-phase waves continue for 40+ seconds.

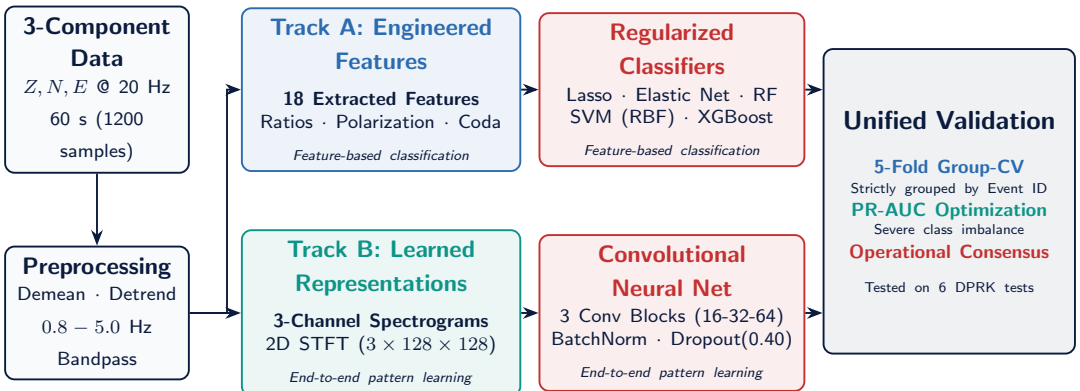
Empirical Discriminant: *Explosions exhibit high vertical-to-horizontal ratios with fast decay; earthquakes show massive horizontal shaking and prolonged coda.*

SECTION 02

Classification Methods

Formulations, loss functions, and regularization across classical ML and deep learning

Two Approaches to Classification



Benchmarking Objective: *Compare manual feature construction against automatic representation learning under identical, strictly event-grouped cross-validation splits.*

Track A: Lasso Logistic Regression (L_1 Regularization)

$$\min_{\mathbf{w},b} \quad -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log \sigma(\mathbf{w}^T\mathbf{x}_i + b) + (1-y_i)\log(1-\sigma(\mathbf{w}^T\mathbf{x}_i + b))\right] \; + \; \lambda\|\mathbf{w}\|_1$$

L_1 Sparsity Mechanism

- **Feature Zeroing:** L_1 penalty ($\|\mathbf{w}\|_1 = \sum |w_j|$) drives redundant feature weights strictly to zero.
- **Physical Selection:** Retains cross-phase spectral ratios while discarding noisy kurtosis metrics.
- **Linear Baseline:** Provides clean, interpretable log-odds ratios for source discrimination.

Tuning & Performance

- **Optimal Parameter:** $C = \frac{1}{\lambda} = 2.0$ (SAGA solver).
- **5-Fold CV Results:**
  - Accuracy: **95.98%** Precision: **98.36%**
  - Recall: **92.58%** PR-AUC: **0.9943**
- **Role:** Fast, highly interpretable linear reference.

Track A: Elastic Net Logistic Regression ($L_1 + L_2$ Penalty)

$$\min_{\mathbf{w},b} \quad \mathcal{L}_{\text{log}}(\mathbf{w}, b) \; + \; \lambda \left[\alpha \|\mathbf{w}\|_1 + \frac{1-\alpha}{2} \|\mathbf{w}\|_2^2 \right], \quad \alpha \; \in \; [0,1] \; (\texttt{l1_ratio})$$

Grouping & Collinearity

- **Overcomes Lasso Limit:** Retains correlated features jointly instead of dropping one at random.
- **Cross-Channel Grouping:** Selects complementary vertical (Z) and horizontal (H) features together.
- **Numerical Stability:** Strictly convex L_2 term stabilizes optimization under collinear ratios.

Tuning & Performance

- **Optimal Parameters:** $C = 2.0$, $\alpha = 0.80$ (80% L_1, 20% L_2).
- **5-Fold CV Results:**
  - Accuracy: **96.27%** Precision: **98.40%**
  - Recall: **94.53%** PR-AUC: **0.9940**
- **Gain over Lasso:** Improves recall by **+1.95%** by preserving multi-channel dynamics.

Track A: Random Forest Classifier (Bagging Ensemble)

Ensemble Aggregation & Split Criterion

$$\hat{f}_{\text{RF}}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x};\Theta_b), \quad I_G(t) = 1 - \sum_{k=0}^{1} p_k^2(t), \quad m = \sqrt{p} \approx 4$$

Ensemble Mechanism

- **Bootstrap Aggregation:** Averages $B = 200$ de-correlated trees to minimize model variance.
- **Subspace Sampling:** Evaluates $m \approx 4$ random features per split, preventing single-feature dominance.
- **Non-Linear Thresholds:** Captures complex wave-property interactions without distribution assumptions.

Tuning & Performance

- **Configuration:** $B = 200$ trees, max_depth $= 8$.
- **5-Fold CV Results:**
  - Accuracy: **96.26%** Precision: **97.39%**
  - Recall: **95.53%** PR-AUC: **0.9936**
- **Robustness:** Tree consensus resists sensor dropouts and noise spikes.

Track A: Support Vector Machine (RBF Kernel)

Dual Formulation & Kernel Function

$$\max_{\boldsymbol{\alpha}} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{N} \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \;\; \text{s.t.} \;\; 0 \leq \alpha_i \leq C, \;\; K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$$

Kernel Margin Separation

- **RBF Projection:** Projects 18D features into Hilbert space for non-linear boundary separation.
- **Support Vector Sparsity:** Decision boundary depends strictly on margin-defining boundary points.
- **Calibrated Confidence:** Platt scaling converts margin distances into calibrated probability scores.

Tuning & Performance

- **Optimal Parameters:** Penalty $C = 20.0$, bandwidth $\gamma = 0.01$.
- **5-Fold CV Results:**
  - Accuracy: **97.13%** (Top Tabular Model)
  - Precision: **97.45%** Recall: **97.05%**
  - PR-AUC: **0.9973**
- **Key Takeaway:** Best classical ML performer, rivaling deep learning accuracy.

Track A: XGBoost Classifier (Gradient Boosted Trees)

Second-Order Objective & Regularized Loss

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{N} \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \gamma T + \frac{\lambda}{2} \sum_{j=1}^{T} w_j^2, \quad g_i = \hat{p}_i - y_i, \quad h_i = \hat{p}_i(1 - \hat{p}_i)$$

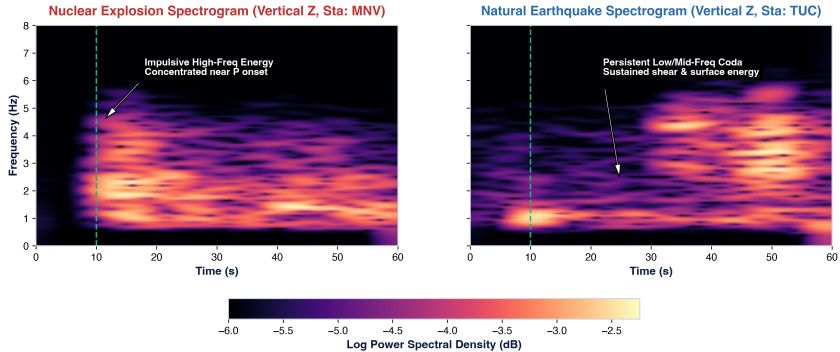
Gradient Tree Boosting

- **Second-Order Curvature:** Uses exact gradients (g_i) and Hessians (h_i) for rapid, precise descent.
- **Complexity Control:** Penalizes total leaves (γT) and leaf weights ($\lambda \|w\|^2$) to prevent overfitting.
- **Shrinkage & Subsampling:** Step size $\eta = 0.05$ with 80% subsampling ensures gradual, robust learning.

Tuning & Performance

- **Configuration:** 200 trees, depth $= 4$, $\eta = 0.05$.
- **5-Fold CV Results:**
  - Accuracy: **96.55%** Precision: **97.89%**
  - Recall: **95.53%** PR-AUC: **0.9954**
- **Physical Feature Gain:** Horizontal spectral spread (`spec_spread_h`) leads with **48.1%** gain.

Track B: Learning Discriminative Patterns from Spectrograms



Explosions

High frequencies concentrated at arrival; rapid temporal decay.

Earthquakes

Prolonged low/mid-frequency energy spanning the entire 60 s record.

CNN Pattern Learning

Learns these 2D patterns directly without manually specified rules.

Track B: 3-Channel 2D Convolutional Neural Network (CNN)

Objective Function: Binary Cross-Entropy with Logits

$$\mathcal{L}_{\text{BCE}}(\mathbf{z}, y) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log \sigma(z_i) + (1 - y_i)\log(1 - \sigma(z_i))\right], \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

CNN Architecture

- **Input Tensor:** $3 \times 128 \times 128$ (Z, N, E spectrograms).
- **4 Conv Blocks:** $3 \to 32 \to 64 \to 128 \to 128$ filters with BatchNorm, ReLU, and MaxPool.
- **Head:** Global pooling $\to$ Dense(128) with Dropout(0.4) $\to$ single output logit.

Training & Performance

- **Training:** AdamW (lr $= 10^{-3}$, decay $= 10^{-2}$), Cosine Annealing.
- **5-Fold CV Results:**
  - Accuracy: **97.98%** (Rank #1 Overall)
  - Precision: **99.01%** Recall: **97.54%**
  - PR-AUC: **0.9976**
- **Advantage:** Directly learns 2D spectral transitions without manual feature engineering.

SECTION 03

Benchmark Results & Feature Importance

Model leaderboard, precision-recall trade-offs, and signal discriminants

Model Leaderboard: 3-Channel CNN Outperforms Classical ML

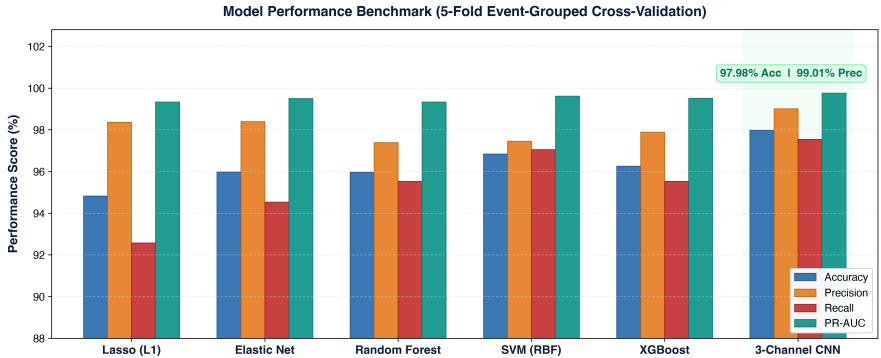
5-Fold Event-Grouped Cross-Validation (Evaluated with zero geographic data leakage)

Rank	Model Family	Accuracy (%)	Precision (%)	Recall (%)	PR-AUC
1	**3-Channel 2D CNN (Spectrograms)**	**97.98**	**99.01**	**97.54**	**0.9976**
2	Support Vector Machine (RBF Kernel)	97.13	97.45	97.05	0.9973
3	XGBoost (Gradient Boosted Trees)	96.55	97.89	95.53	0.9954
4	Elastic Net Logistic Regression	96.27	98.40	94.53	0.9940
5	Random Forest	96.26	97.39	95.53	0.9936
6	Lasso Logistic Regression (L_1)	95.98	98.36	92.58	0.9943

CNN Representation Advantage: Learning directly from 2D spectrograms captures joint time-frequency patterns that summary features miss, leading across all metrics.

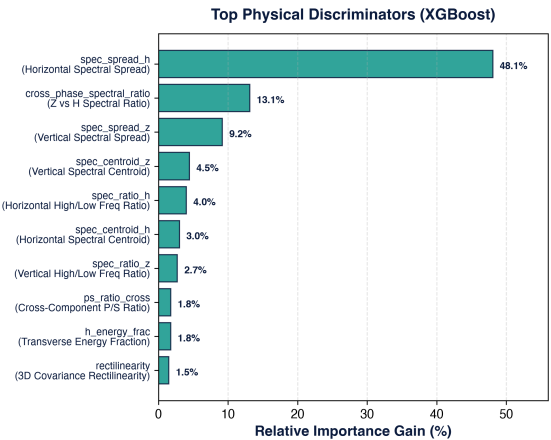
Strong Tabular Baselines: Non-linear SVM and tree models achieve $> 96\%$ accuracy, proving that engineered energy ratios and frequency metrics capture substantial signal.

Performance Benchmark: Multi-Metric Model Comparison



Key Finding: *All 6 models achieve $> 94\%$ accuracy; the 3-Channel 2D CNN leads across all evaluation metrics (97.98% Acc, 99.01% Prec, 99.76% PR-AUC).*

Feature Importance: What Drives Model Decisions?



48.1% Horizontal Spread

Earthquakes produce broad horizontal shear motion; explosions show narrow, concentrated horizontal signals.

13.1% Vertical-to-Horizontal Ratio

Explosions are dominated by initial vertical compressional motion; earthquakes produce substantial horizontal energy.

9.2% Vertical High Frequencies

Compact blast sources concentrate energy at higher frequencies than large fault ruptures.

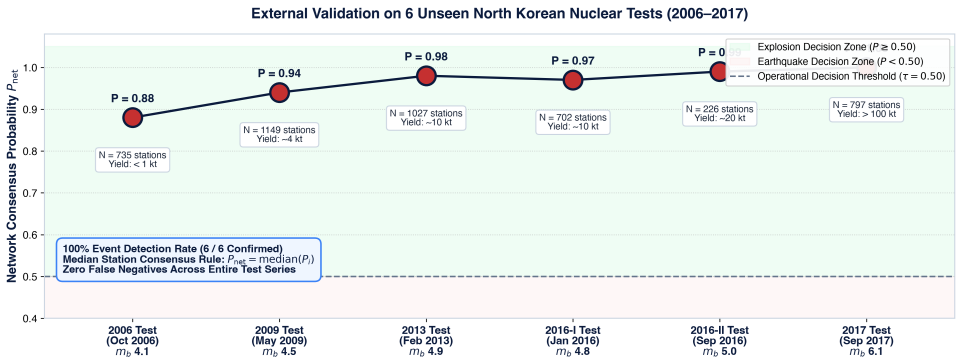
SECTION 04

External Generalization: North Korea (2006–2017)

Testing model generalization against six declared DPRK underground nuclear tests

Note: DPRK stands for the Democratic People's Republic of Korea (North Korea).

External Validation: 100% Detection Across All 6 DPRK Tests



Zero False Negatives: *All 6 unseen nuclear tests classified with high confidence ($P_{\text{net}} \geq 0.88$). These events were completely withheld from model training and hyperparameter tuning.*

Operational Reality: Multi-Station Consensus & Class Imbalance

Transitioning from single-sensor models to operational network monitoring.

Multi-Station Median Consensus

- Single sensor traces suffer from local noise and signal attenuation.
- Global monitoring networks record each event across hundreds of stations ($N = 226$ to $1,149$).
- **Network Decision:** $P_{\text{net}} = \text{median}_i(P_i)$.
- Aggregation cancels uncorrelated local noise, driving robust $\mathbf{100}\%$ event detection across all tests.

The Class Imbalance Challenge

- Natural earthquakes outnumber nuclear tests by $> 100,000 : 1$.
- In extreme imbalance, a 1% single-station false-positive rate yields frequent false alarms:
  $P(\text{Blast} \mid +) = \frac{\text{TPR} \cdot \pi}{\text{TPR} \cdot \pi + \text{FPR} \cdot (1-\pi)}$
- Requiring multi-station network consensus pushes false alarms toward zero, enabling reliable monitoring.

SUMMARY & CONCLUSIONS

Key Takeaways for Statistical Monitoring

1. Feature Engineering

Combining 18 energy ratios and spectral decay metrics yields $> 96\%$ accuracy with classical classifiers while strictly preventing geographic data leakage.

2. Representation Learning

2D STFT spectrograms allow CNNs to learn directly from time-frequency images without manual thresholds, leading the benchmark (97.98% Acc, 0.9976 PR-AUC).

3. Operational Consensus

Multi-station median voting overcomes extreme natural class imbalance, successfully identifying all 6 North Korean nuclear tests with zero false negatives.

Thank You